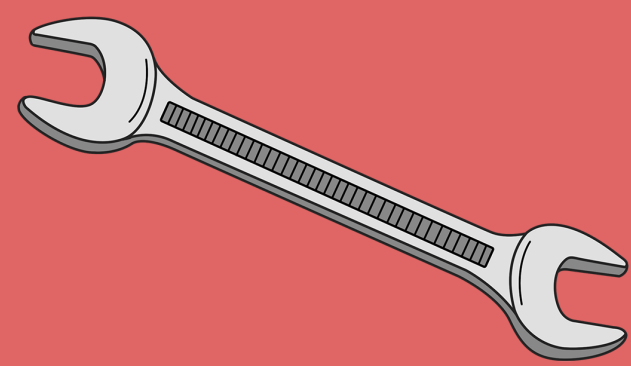




Diagnosing Tool-Call Decision Making in Small Language Models

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Abstract

Tool-calling language models must decide **whether to call a tool, ask for missing information, or state that the available tools are insufficient**. We study this three-way decision problem in small language models using **Gemma-3-4B-it** and **Llama-3.2-3B-Instruct**. We compare **prompting, linear probes over hidden states, supervised fine-tuning (SFT), Direct Preference Optimization (DPO), and a Constitutional AI (CAI)-style pipeline** based on rule-guided self-critique, self-revision, and preference optimization.

Introduction

Small Tool-calling Language Models may be brittle when deciding **whether tool use is appropriate** before making a tool call.

We study this problem using the **When2Call** dataset, where each example has a corresponding label of **tool call, request for info, or cannot answer**.

Our primary investigation is centered around **whether small language models are able to learn this decision boundary** through techniques such as prompting, post-training, and a synthetic Constitutional AI-style supervision. We then investigate: **are failures caused due to a lack of knowledge, or if this decision is represented in model embeddings but not expressed in the final answer?**.



Methodology

We evaluate tool-calling as a **three-way classification problem**: given a user request and a set of available tools the model must choose between **tool_call**, **request_for_info**, or **cannot_answer** classes. For example, if a weather tool is available and the user asks for the weather in College Park, the correct action is **tool_call**. If the user asks "What is the weather there?" without giving a location, the model should choose **request_for_info**. If the user asks for legal advice and no available tool can answer it, the correct action is **cannot_answer**.

Our intervention types:

-Prompting: We test zero-shot and four-shot prompts to measure whether models can follow decision rules from instructions and examples alone.

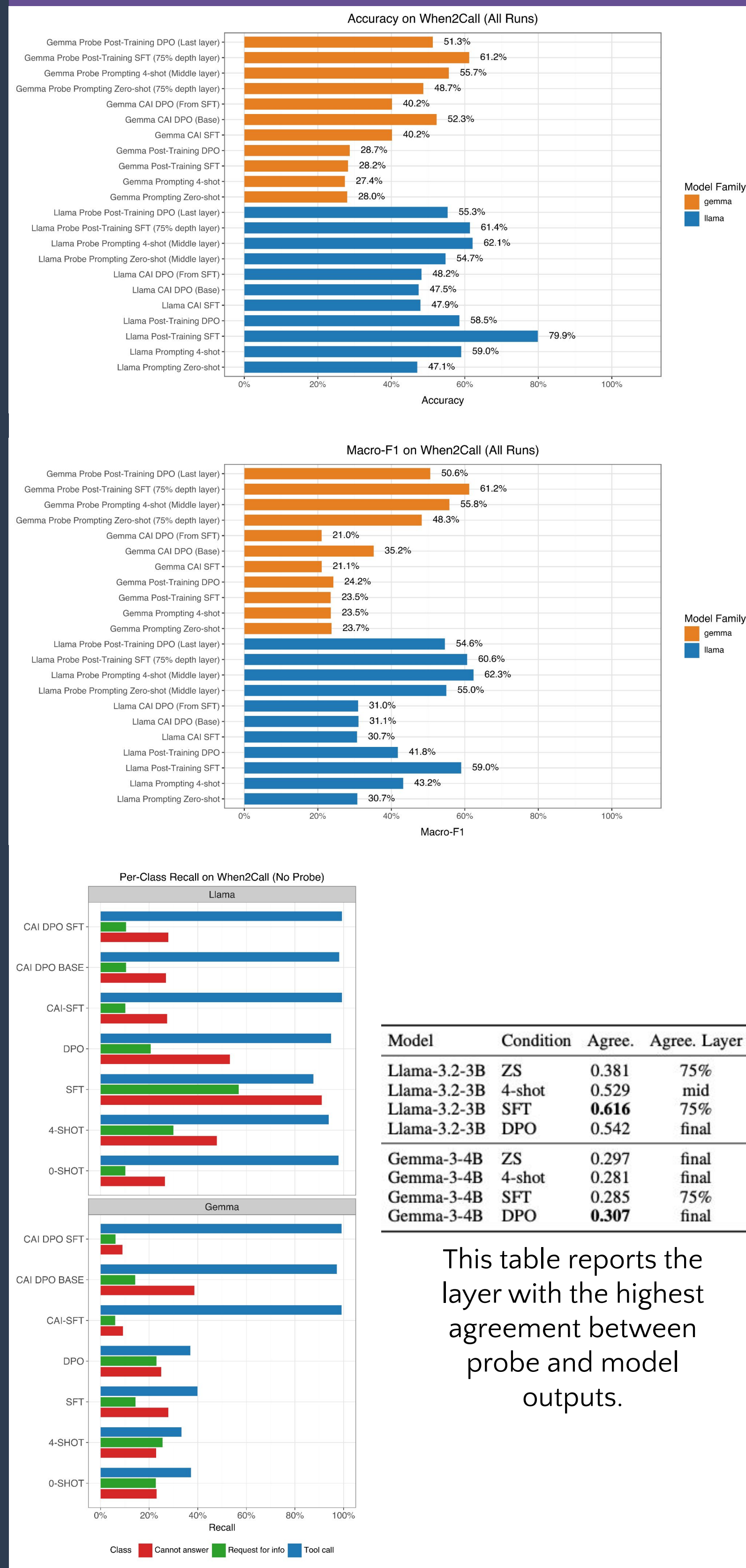
-Post-training: We fine-tune Gemma-3-4B-it and Llama-3.2-3B-Instruct using supervised fine-tuning and direct preference optimization on When2Call training data.

-CAI-style supervision: We write a task-specific tool-calling constitution that tells the model when to call a tool, ask for missing fields, or decline because the tools are insufficient. In the weather example, this means the constitution should push revisions and preferences toward calling the weather tool when a location is given, but asking for the missing location when it is not. This tests whether rule-guided synthetic supervision can reduce reliance on expensive human-labeled data.

-Linear probing: We extract hidden states from the final prompt token and train logistic regression probes to predict the decision label. We probe the middle layer, 75% layer, and final layer. This tests whether the correct decision is internally recoverable even when the final model output is wrong.

We evaluate on the **When2Call multiple-choice test set** and report **accuracy, macro-F1, class-wise behavior, prediction distributions, and probe accuracy**.

Results



Analysis/Discussion

Llama responds best to direct supervised targets: standard **SFT reaches 0.799 accuracy and 0.590 macro-F1**, suggesting that labeled examples align its final action policy well. Gemma is harder to reshape with standard SFT or DPO, but CAI-DPO from the base model improves accuracy from **0.280 to 0.523** and macro-F1 from **0.237 to 0.352**.

However, **CAI's gains are very uneven**: its improvements **concentrate on tool_call**, while **request_for_info** and **cannot_answer** remain difficult. Qualitative examples show why: CAI revisions often make responses look more **tool-call-like instead of preserving the correct decision class**. In the weather example, this means fabricating a missing location rather than asking the user for it.

Linear probes reveal that the correct decision is often **recoverable from hidden states** even when the model's final output is wrong. This suggests a representation-behavior gap, where small models may encode decision-relevant information but fail to express it reliably.

Overall, better tool-call decision making requires both **stronger alignment between internal signals and final behavior**, and **stricter verification that synthetic CAI targets remain decision-faithful**.

Conclusion/Future Work

Reliable tool use requires deciding **when not to use tools**. Our study shows that **this decision boundary is fragile in small language model**: some errors occur because recoverable decision information is not expressed in the final output, while others arise when synthetic supervision fails to preserve the intended action class. The main challenge is not just teaching models how to call APIs, but **teaching them to distinguish between available, underspecified, and insufficient tool settings**. Overall, small model tool use needs training methods that preserve the intended decision class and **avoid turning every ambiguous or unsupported request into a tool call**. Future systems could **separate decision making from tool-call formatting** and **verify synthetic supervision before training**.